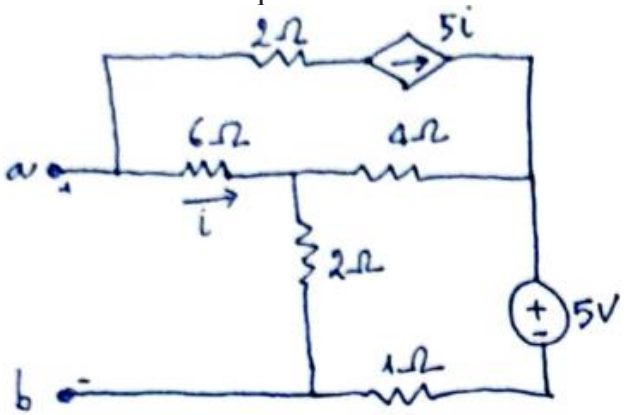
HCMC UNIVERSITY OF TECHNOLOGY AND EDUCATION
Faculty for High Quality Training

Program name: Power Electrical Engineering

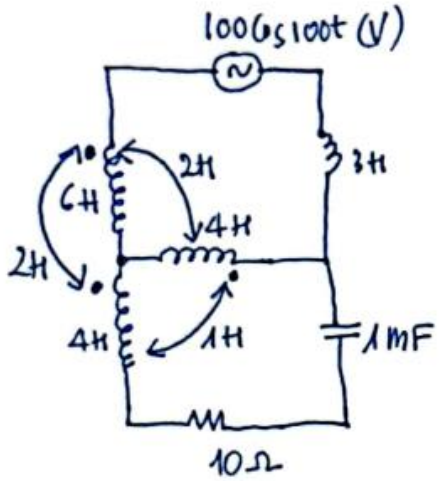
Control and Automation

FINAL EXAMINATION
SEMESTER 1 – ACADEMIC YEAR 2021-2022
Course name: Electrical Circuit
Course ID: ELCI140144E
Exam code: EC02 Number of pages: 03
Duration: 90 minutes.
opened books.

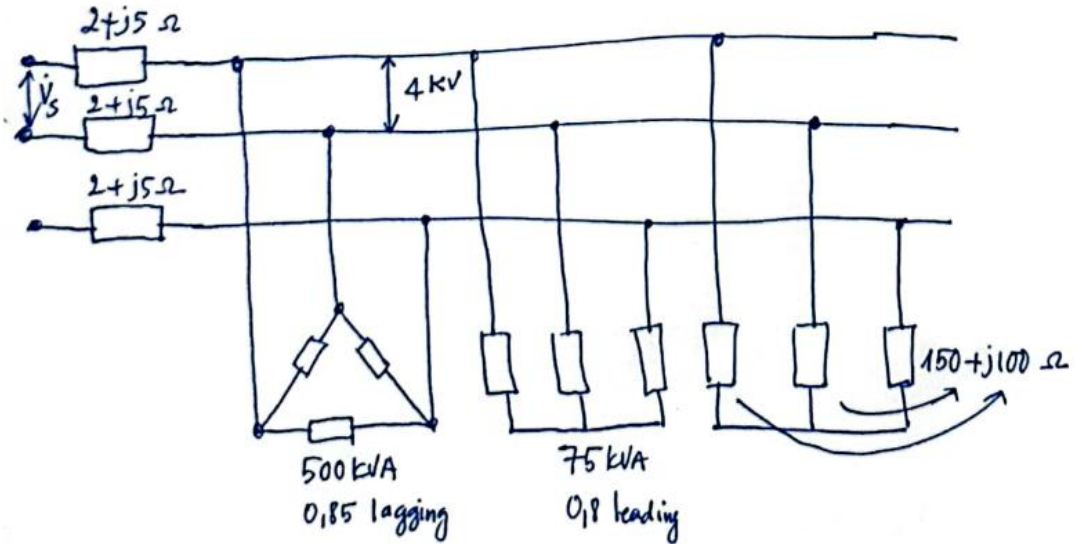
Question 1: (1.5/10) Find the Thevenin equivalent circuit for the circuit below



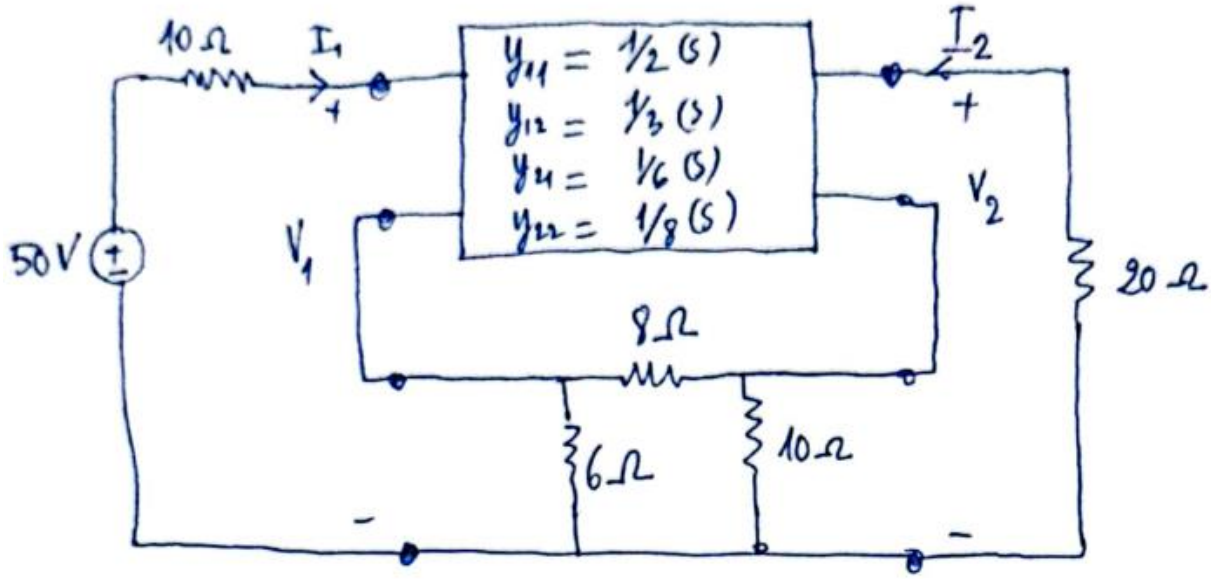
Question 2: (1.5/10) Determine the complex power supplied by the source in the circuit shown in Figure below



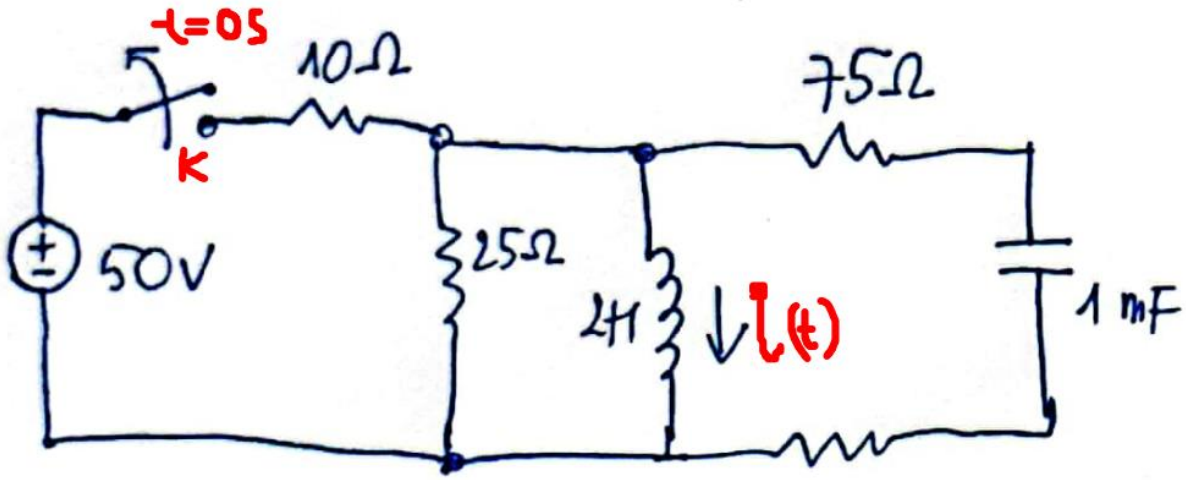
Question 3: (1.0/10)
Find V_s in the figure below.



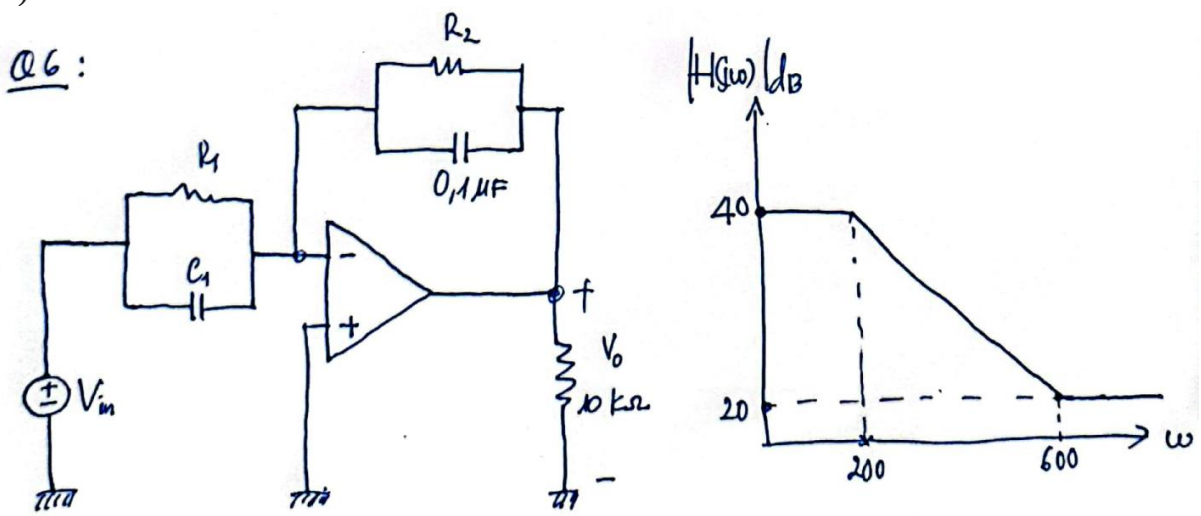
Question 4: (2.0/10)
Find Z parameters of the combined two-port network in the figure below. Find the power absorbed by 20 ohm resistor.



Question 5: (2.0/10)
At t = 0 s, K is opened. Find i(t) in the figure below.



Question 6: (2.0/10)
Design the circuit shown in Figure (a) to have the asymptotic Bode plot shown in Figure (b).



Note: Proctors are not allowed to give any unauthorized explanation.

CLOs	Description	Assessed in
G1.1	Ability to describe and identify the electric circuit and the basic elements of the electric circuit	Questions 1,2,3,4,5,6
G1.2	the ability to apply the Kirchoff's law, equivalent transformation for solving the circuit problems. Determine the supply power, dissipation power, and the power preservation in a circuit.	Questions 1, 2, 3, 4 ,5,6
G1.3	Ability to analyze circuits using the node-voltage method, the mesh-current method, Thevenin and Norton equivalents, and superposition property.	Questions 1,3,4,5,6
G1.4	Ability to understand and solve AC circuits one phase and three phases	Questions 1,2,3
G1.5	Ability to determine the parameters of two-port network	Questions 4
G1.6	Ability to analyze and calculate the values of the transient responses and sketch waveforms of the currents and drop voltages in the time domain	Questions 5
G1.7	Ability to analyses and calculate the current and voltage in the time responses with non-sinusoidal source and Bode plots.	Questions 6

Jan 01th, 2022

Approved by program chair